Abbas Ali

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PhD in Electrical and Computer Engineering • Anticipated Graduation: December 2029

RESEARCH SUMMARY

PhD researcher at the intersection of FPGA design and hardware security for cyber-physical systems, developing hardware fingerprinting for physical-layer device authentication that identifies embedded devices from the analog imperfections in their digital switching signals. Validated on real production automotive ECUs and extending to drones and robotic platforms, since the underlying device physics is common to all embedded hardware. Two years of industry experience in FPGA and RTL design before the PhD: SystemVerilog, VHDL, and Verilog; SoC integration across AXI, DMA, and the PS/PL boundary into DDR; synthesis and timing closure; and the debugging that begins where simulation stops matching the board. Solves hardware constraints by reformulating the algorithm rather than spending area on it.

EDUCATION

PhD in Electrical and Computer Engineering — January 2025 – **December 2029 (anticipated)**
University of Michigan-Dearborn, MI — *Advisor: Prof. Hafiz Malik, ISSF Laboratory*
Coursework: Computer Architecture, Reconfigurable Computing, Embedded Systems, Deep Learning, Pattern Recognition

BS in Electrical Engineering — September 2018 – August 2022
University of Engineering & Technology (UET), Peshawar, Pakistan — *CGPA 3.83/4.00, Gold Medalist*

PUBLICATIONS

Peer-Reviewed

[1] **A. Ali** and H. Malik, *"Short: Is Your ECU Really Yours?"* 4th USENIX Symposium on Vehicle Security and Privacy (VehicleSec '26), Baltimore, MD, August 2026. **Artifact Evaluated: Available, Functional, Reproduced.** [Link]

[2] H. Arif, **A. Ali**, S. Kiyani, and H. Malik, *"Real-Time Authentication of Embedded Hardware,"* 4th EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP '26), Detroit, MI, 2026. **Accepted.**

[3] **A. Ali**, A. Shah, A. H. Khan et al., *"Speed vs. Efficiency: A Framework for High-Frequency Trading Algorithms on FPGA using Zynq SoC," Alexandria Engineering Journal*, vol. 96, pp. 1–14, 2024. [DOI]

[4] A. W. Mohammed, **A. Ali**, H. Arif, F. R. P. Mohammed, and H. Malik, *"A Lightweight IDS Framework Using FPGA-Based Hardware Fingerprinting on Zynq SoC,"* IEEE International Conference on Cyber Security and Resilience (CSR), Chania, Greece, pp. 544–549, 2025. [DOI]

[5] A. H. Khan, A. Shah, **A. Ali** et al., *"A Performance Comparison of ML Models for Stock Market Prediction with a Novel Investment Strategy," PLOS ONE*, vol. 18, no. 9, e0286362, 2023. [DOI]

RESEARCH EXPERIENCE

Graduate Research Assistant, Hardware Security and FPGA Systems — January 2025 – Present
Information Systems, Security, and Forensics (ISSF) Laboratory, University of Michigan-Dearborn (NSF-Funded)

- Developed physical-layer fingerprinting that identifies devices from their switching transients, detecting clones that pass cryptographic checks with **99.5%** real-time accuracy. Published at USENIX VehicleSec '26.
- Validated on **9 electrically identical production GCM 048 ECUs from New Eagle** across 13 sessions over several weeks, showing the fingerprint holds over time.
- Built a real-time on-board authentication pipeline: 125 MS/s DMA capture on Red Pitaya STEMlab, waveform feature extraction, and ARM32 Docker inference with no host PC.
- Ported acquisition to Zynq-7000 with the AD9467-FMC ADC, moving capture into PL with DMA into PS DDR, and held 100% accuracy across all ADC and probe combinations. Published at IEEE CSR 2025.
- Extending the method to drones, robotic arms, GPUs, battery management systems, and hardware-in-the-loop test stations.

PROFESSIONAL EXPERIENCE

Lead FPGA Engineer — September 2022 – October 2024
IntelliFPGA US

- Designed the first documented FPGA architectures for the Aroon, MACD, and RSI trading indicators on Zynq SoC, replacing all six dividers with constant multiply and shift operations and pipelining the datapath for a **30x** speedup over software.
- Built the AD9467 ADC and AD9517 clock acquisition path end to end: configuration firmware, PL capture, PS/PL transfer into DDR, and real-time Ethernet streaming.
- Built custom AXI-Lite and AXI-Stream IP with DMA to move photon timestamps from PL to PS, compressing each from 32 to

24 bits to keep **25% more detections** within the 20 MB/s SD card limit.
- Migrated a legacy Altera codebase to VHDL on Intel Stratix 10 SoC, rebuilding DMA, AXI, and IP blocks, verified in ModelSim and Questa and debugged on hardware with ILA.

Embedded Systems Engineer (Intern) September 2021 – October 2021
Chip Soul Technology, Peshawar, Pakistan. Wrote C/C++ firmware for the STC89C52 microcontroller. Implemented I2C, SPI, EPROM, and RTC peripheral interfaces.

TALKS AND PRESENTATIONS

- **Conference Talk**, *"Short: Is Your ECU Really Yours?"* USENIX VehicleSec '26, Baltimore, MD, August 2026
- **Lightning Talk**, *"Toward Physical-Layer Fingerprinting of Production Automotive ECUs via CAN Bus Signals."* USENIX VehicleSec '26, August 2026
- **Live Hardware Demonstration**, USENIX VehicleSec '26, August 2026
- **Poster (1st Place)**, *"A Lightweight IDS Framework Using FPGA-Based Hardware Fingerprinting on Zynq SoC."* CCAT Global Symposium on Mobility Innovation, Ann Arbor, MI, April 2026. With S. Teja Jarabala and S. Kumbham

SELECTED HARDWARE PROJECTS

16-bit Microprocessor in VHDL (ECE 575): Single-cycle 16-bit CPU on Zynq-7020. Full datapath with ALU, register file, instruction memory, and control unit. Functionally verified, timing closed at 143 MHz.

CNN Accelerator on PYNQ-Z2 (ECE 554): Hardware and software co-design from PyTorch training (95.80% accuracy, 43-class GTSRB) through Vitis HLS synthesis, INT8 quantization, and AXI deployment. 89% hardware accuracy at 101 MHz on 55% of available LUTs.

MNIST Neural Network in RTL (ECE 5752): Pipelined MAC engines in VHDL for a fully-connected network. Fixed-point weights in BRAM, deployed on Zynq-7020 over AXI-Stream and ARM DMA.

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, VHDL, C/C++, Python, Tcl, MATLAB
Design and Verification: RTL design, AXI, DMA, PS/PL, testbenches, static timing analysis, timing closure
Tools and Hardware: Vivado, Vitis HLS, ModelSim, Questa, ILA, Linux, Git, Zynq-7000, Stratix 10, Red Pitaya, PicoScope

HONORS

- **1st Place, Poster Competition**, 2026 CCAT Global Symposium on Mobility Innovation, Ann Arbor, MI
- **Gold Medalist**, UET Peshawar, top of Electrical Engineering graduating class
- **Teaching Assistant**, FPGA Design and PCB Workshop, UET Peshawar